

Comets

Comets are fragile balls of snow and dust that live at the edge of the solar system in a vast cloud known as the Oort Cloud. At the center of the comet is the nucleus—a dirty ball of snow that measures a few miles across. If a comet passes too close to the Sun, the snow turns into gas, releasing dust and gas in the process. This forms a vast cloud of material called a coma, consisting of a huge spherical head and one or two tails.



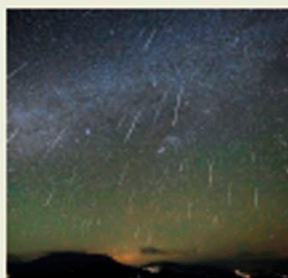
Periodic comets

When a comet leaves the Oort Cloud, it can travel on an orbit that brings it close to the Sun at regular intervals. Short-period comets, such as Halley's Comet, take less than 200 years to orbit the Sun.

Serial no.	Name	Orbital period	Sightings	Next due
1	1P/Halley	75 years	30	July 2061
2	2P/Encke	3 years, 3 months	62	March 2017
3	6P/d'Arrest	6 years, 5 months	20	September 2021
4	9P/Tempel	5 years, 5 months	12	January 2022
5	17P/Holmes	6 years, 8 months	10	February 2021
6	21P/Giacobini-Zinner	6 years, 6 months	15	September 2018
7	29P/Schwassmann-Wachmann	15 years	7	March 2019
8	39P/Oterma	19 years	4	July 2023
9	46P/Wirtanen	5 years, 4 months	10	December 2018
10	50P/Arend	8 years, 2 months	8	April 2024
11	55P/Tempel-Tuttle	33 years	5	May 2031
12	67P/Churyumov-Gerasimenko	6 years, 4 months	7	December 2021
13	81P/Wild	6 years, 4 months	6	November 2022
14	109P/Swift-Tuttle	133 years	5	July 2126

Meteors

As comets and asteroids travel through space, they shed lumps of rock and dust called meteoroids. The smallest meteoroids burn up as they pass through Earth's atmosphere, producing streaks of light known as meteors. Most meteors glow for only a few seconds before burning up. If the meteor does not burn up completely, the remaining piece lands on Earth's surface and is known as a meteorite. Most meteorites that land on Earth are no bigger than a small rock.



Meteor showers

When Earth passes through a concentration of meteoroids, it produces a shower of meteors, or "shooting stars". Below is a list of some of the major meteor showers with the dates they are most likely to occur each year.

Serial no.	Name	Peak date	Most meteors	Parent comet/asteroid
1	Quadrantids	January 4	120 per hour	2003 EH1
2	Lyrids	April 22	10 per hour	C/1861 G1 (Thatcher)
3	Eta Aquarids	May 5	30 per hour	1P/Halley
4	Perseids	August 12	100 per hour	109P/Swift-Tuttle
5	Geminids	December 14	120 per hour	3200 Phaethon